Volatility Transmission Between Western and Asian Equity Markets

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1 Introduction

This report studies how volatility is transmitted between Western and Asian equity markets and how the strength of these links changes over time. The empirical analysis focuses on five benchmarks: the United States (SP 500), the Euro Area (Euro Stoxx 50), Japan (Nikkei 225), China (Hang Seng), and India (Nifty 50). Our expectations are that connectedness increases during crisis episodes and that Western markets, especially the United States, tend to play a leading role in transmitting volatility.

The analysis follows a two-step approach. First, conditional volatilities are estimated from equity returns using GARCH models. Since equity returns typically display heavy tails and asymmetric reactions to negative shocks, we compare Gaussian and Student-t distribution and allow for asymmetric volatility dynamics through EGARCH and GJR specifications. This step produces one conditional volatility series for each market.

Second, the estimated volatility are modeled jointly in a vector autoregression (VAR). Volatility spillovers are measured using the Diebold–Yilmaz framework, which is based on generalized forecast error variance decompositions. The main outputs are: (i) a total connectedness index, (ii) directional measures that show how much volatility of each market transmits to and receives from the rest, and (iii) net spillovers that classify markets as volatility givers or receivers.

2 Methodology

2.1 Data

The empirical analysis uses five equity market benchmarks: the S&P 500 (United States), the Euro Stoxx 50 (Euro Area), the Nikkei 225 (Japan), the Hang Seng (China proxy), and the Nifty 50 (India). We collect daily closing prices from Yahoo Finance and construct log weekly returns.

$$r_{i,t} = 100 \times \log\left(\frac{P_{i,t}^{(W)}}{P_{i,t-1}^{(W)}}\right) \tag{1}$$

The analysis is carried out at weekly frequency to reduce distortions due to non-synchro trading cause time-zone differences across regions, which can affect the direction of spillovers when using daily returns.

2.2 Conditional volatility estimation

For each market i, we estimate conditional volatility using GARCH models. Let

$$r_{i,t} = \mu_i + \varepsilon_{i,t}, \qquad \varepsilon_{i,t} = \sigma_{i,t} z_{i,t}, \tag{2}$$

where $\sigma_{i,t}^2$ is the conditional variance and $z_{i,t}$ is an i.i.d. innovation. We consider two distributional assumptions for $z_{i,t}$: Gaussian and Student-t

GARCH(1,1)

$$\sigma_{i,t}^2 = \omega_i + \alpha_i \varepsilon_{i,t-1}^2 + \beta_i \sigma_{i,t-1}^2, \tag{3}$$

with $\omega_i > 0$, $\alpha_i \geq 0$, $\beta_i \geq 0$.

GJR-GARCH(1,1) To capture asymmetric volatility responses to negative shocks, we also estimate the GJR-GARCH(1,1) model

$$\sigma_{i,t}^2 = \omega_i + \alpha_i \varepsilon_{i,t-1}^2 + \gamma_i \mathbf{1}\{\varepsilon_{i,t-1} < 0\} \varepsilon_{i,t-1}^2 + \beta_i \sigma_{i,t-1}^2, \tag{4}$$

where $\gamma_i > 0$ implies that negative shocks have a larger impact on future volatility than same positive shocks.

EGARCH(1,1) As an alternative asymmetric model, we estimate EGARCH(1,1),

$$\log(\sigma_{i,t}^2) = \omega_i + \beta_i \log(\sigma_{i,t-1}^2) + \alpha_i \left(|z_{i,t-1}| - E|z_{i,t-1}|\right) + \gamma_i z_{i,t-1}. \tag{5}$$

This specification models $\log(\sigma_{i,t}^2)$ and therefore does not require non-negativity constraints on variance parameters.

2.3 VAR model for volatilities

Let $\hat{\sigma}_t = (\hat{\sigma}_{1,t}, \ldots, \hat{\sigma}_{N,t})'$ denote the vector of estimated conditional volatilities across markets obtained from the selected univariate GARCH models. We model $\hat{\sigma}_t$ using a VAR(p),

$$\hat{\sigma}_t = c + \sum_{k=1}^{p} A_k \hat{\sigma}_{t-k} + u_t, \qquad u_t \sim (0, \Sigma_u), \tag{6}$$

where c is an $N \times 1$ intercept, A_k are $N \times N$ coefficient matrices, and Σ_u is the reduced-form covariance matrix.

If the VAR is stable (all eigenvalues inside the unit circle), it admits a Wold representation,

$$\hat{\sigma}_t = \mu + \sum_{h=0}^{\infty} \Psi_h u_{t-h}, \tag{7}$$

where $\Psi_0 = I_N$ and the matrices Ψ_h are obtained recursively from the estimated VAR coefficients. This MA representation is the input for the forecast error variance decomposition used in the connectedness analysis.

2.4 GFEVD and Diebold–Yilmaz

Connectedness is computed from the generalized FEVD (order-invariant, no Cholesky orthogonalization). For horizon H, define

$$\theta_{ij}^{(g)}(H) = \frac{\sigma_{jj}^{-1} \sum_{h=0}^{H-1} \left(e_i' \Psi_h \Sigma_u e_j\right)^2}{\sum_{h=0}^{H-1} e_i' \Psi_h \Sigma_u \Psi_h' e_i}, \qquad \tilde{\theta}_{ij}^{(g)}(H) = \frac{\theta_{ij}^{(g)}(H)}{\sum_{j=1}^{N} \theta_{ij}^{(g)}(H)}.$$

Using $\tilde{\Theta}(H) = \{\tilde{\theta}_{ij}^{(g)}(H)\}$, the Diebold–Yilmaz indices are:

$$C(H) = 100 \frac{\sum_{i \neq j} \tilde{\theta}_{ij}^{(g)}(H)}{N}, \qquad C_{i \leftarrow \bullet}(H) = 100 \sum_{j \neq i} \tilde{\theta}_{ij}^{(g)}(H),$$

$$C_{i \rightarrow \bullet}(H) = 100 \sum_{j \neq i} \tilde{\theta}_{ji}^{(g)}(H), \qquad C_i^{net}(H) = C_{i \rightarrow \bullet}(H) - C_{i \leftarrow \bullet}(H).$$

Here e_i is the i-th selection vector, σ_{jj} is the j-th diagonal element of Σ_u, and Ψ_h comes from the VAR moving-average representation. Positive $C_i^{net}(H)$ identifies net transmitters, negative values net receivers.

3 Empirics

3.1 Evidence on returns

The table reports descriptive statistics for weekly log-returns. The sample includes 954 observations for each market. All series display negative skewness, meaning that large negative returns are more likely than large positive returns. In addition, kurtosis is high across markets (these are kurtosis values not excess kurtosis), value above 3 indicate heavy tails.

These unconditional features motivate us to compare Gaussian and Student-t distributions. However, kurtosis and skewness are also affected by time-varying volatility. Therefore, the final choice between Gaussian and Student-t distributions is confirmed by comparing likelihood, AIC, BIC.

Table 1: Descriptive statistics for weekly log-returns

Ticker	Mean	Std	Skewness	Kurtosis
^GSPC	0.159182	2.502789	-0.921819	10.447037
^HSI	0.001562	3.114327	-0.118847	5.929595
^N225	0.121399	3.068949	-1.049350	12.493855
^NSEI	0.174993	2.715406	-0.656355	9.648077
^STOXX50E	0.033176	3.042806	-1.317807	12.306930

3.2 Volatility model estimation and selection

In this part conditional volatility is estimated separately for each market using six candidate models: GARCH(1,1), EGARCH(1,1), and GJR-GARCH(1,1), each estimated under Gaussian and Student-t. We compare their results in terms of Likelihood, AIC, BIC.

The comparison is reported market by market in Tables 2-6. We can see that across the markets, the best model is GJR-GARCH(1,1) with Student-t. It delivers the highest log-likelihood and the lowest AIC and BIC. This choice is consistent with the return evidence.

Table 2: Volatility model comparison for the SP500

Model	LogLik	k	AIC	BIC
GJR(1,1)-t	-2022.418034	6	4056.836068	4086.000050
GARCH(1,1)-t	-2044.499794	5	4098.999589	4123.302907
EGARCH(1,1)-t	-2045.262065	5	4100.524131	4124.827449
GJR(1,1)	-2050.990974	5	4111.981947	4136.285266
EGARCH(1,1)	-2071.074614	4	4150.149229	4169.591884
GARCH(1,1)	-2072.995254	4	4153.990508	4173.433163

Table 3: Volatility model comparison for the Euro Stock 50

Model	LogLik	k	AIC	BIC
GJR(1,1)-t	-2247.127598	6	4506.255197	4535.419179
EGARCH(1,1)-t	-2258.809097	5	4527.618193	4551.921512
GARCH(1,1)-t	-2259.516046	5	4529.032091	4553.335409
GJR(1,1)	-2289.657098	5	4589.314197	4613.617515
EGARCH(1,1)	-2308.227090	4	4624.454180	4643.896835
GARCH(1,1)	-2313.101708	4	4634.203416	4653.646071

Table 4: Volatility model comparison for the Nikkei 225

Model	LogLik	k	AIC	BIC
GJR(1,1)-t	-2309.690642	6	4631.381285	4660.545267
GARCH(1,1)-t	-2319.887299	5	4649.774598	4674.077916
EGARCH(1,1)-t	-2321.561410	5	4653.122819	4677.426138
GJR(1,1)	-2324.053174	5	4658.106348	4682.409666
GARCH(1,1)	-2347.423526	4	4702.847052	4722.289707
EGARCH(1,1)	-2349.776638	4	4707.553276	4726.995930

Table 5: Volatility model comparison for the Hang Seng

Model	LogLik	k	AIC	BIC
GJR(1,1)-t	-2359.344337	6	4730.688674	4759.852656
GARCH(1,1)-t	-2366.902424	5	4743.804847	4768.108166
EGARCH(1,1)-t	-2367.512184	5	4745.024367	4769.327686
GJR(1,1)	-2371.831657	5	4753.663314	4777.966632
EGARCH(1,1)	-2379.436948	4	4766.873895	4786.316550
GARCH(1,1)	-2379.771513	4	4767.543026	4786.985681

Table 6: Volatility model comparison for the Nifty 50

Model	LogLik	k	AIC	BIC
GJR(1,1)-t	-2130.428328	6	4272.856656	4302.020638
GARCH(1,1)-t	-2140.558239	5	4291.116478	4315.419796
GJR(1,1)	-2140.603875	5	4291.207750	4315.511069
EGARCH(1,1)-t	-2143.076505	5	4296.153010	4320.456328
GARCH(1,1)	-2154.747797	4	4317.495593	4336.938248
EGARCH(1,1)	-2160.818670	4	4329.637341	4349.079996

3.3 Estimated conditional volatility series

Using the selected GJR-t model, we estimate conditional volatility series for each benchmarks. Figure 1 plots the resulting series that show the typical features of financial volatility: volatility clustering and pronounced spikes during periods of market stress.

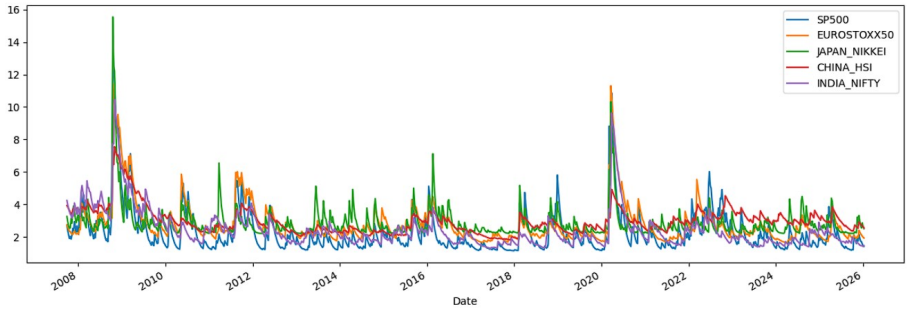


Figure 1: Estimated weekly conditional volatility series.

3.4 VAR specification

In this step, we model the vector of estimated conditional volatilities, $\sigma_t = (\sigma_{1,t}, \ldots, \sigma_{N,t})'$, using a VAR($p$). Table 7 reports AIC and BIC for candidate orders: BIC is minimized at $p = 1$, while AIC reaches its minimum at $p = 4$. We adopt $p = 1$ as the baseline specification because BIC applies a stronger penalty for additional parameters.

Table 7: Lag-order selection for the VAR: information criteria.

p	AIC	BIC
1	-12.331557	-12.178579
2	-12.374626	-12.093931
3	-12.437200	-12.028577
4	-12.444750	-11.907983
5	-12.416967	-11.751843
6	-12.397236	-11.603539
7	-12.408501	-11.486016
8	-12.384260	-11.332770
9	-12.365004	-11.184292
10	-12.336769	-11.026618
11	-12.305450	-10.865642
12	-12.275402	-10.705718

Then we verify VAR stability. The estimated VAR(1) is stable as we can see in Table 8; the eigenvalues of the companion matrix are all strictly below one in modulus.

Table 8: Stability for the estimated VAR(1)

Diagnostic	Value
Eigenvalue 1	0.7346
Eigenvalue 2	0.9618
Eigenvalue 3	0.8679
Eigenvalue 4	0.8679
Eigenvalue 5	0.8837

The estimated VAR(1) shows strong persistence in all volatility series. The own-lag coefficients are large and highly significant (0.93 for the S&P 500, 0.82 for the Euro Stoxx 50, 0.72 for the Nikkei 225, 0.95 for the Hang Seng, and 0.89 for the Nifty 50), indicating substantial volatility persistence at weekly frequency.

Table 9: VAR(1) summary

Asset	Own-lag coefficient	Significance
SP500	0.930	***
ESTOXX50	0.820	***
NIKKEI	0.724	***
HSI	0.947	***
NIFTY	0.893	***

From the table 10 we can see that the highest correlations are among the developed markets (S&P500–Euro Stoxx 50 and Euro Stoxx 50–Nikkei 225), while China and India remain positively correlated with the rest of the system. The residuals are contemporaneously correlated across markets, implying that VAR shocks cannot be treated as independent. For this reason, we use the generalized FEVD within the Diebold–Yilmaz framework. This approach is appropriate when the error covariance matrix Σ is not diagonal and delivers spillover measures that are invariant to the ordering of variables in the VAR, unlike Cholesky decompositions (orthogonalized errors).

Table 10: Residual correlation matrix from the estimated VAR(1).

	SP500	ESTOXX50	NIKKEI	HSI	NIFTY
SP500	1.000	0.738	0.654	0.552	0.568
ESTOXX50	0.738	1.000	0.735	0.590	0.549
NIKKEI	0.654	0.735	1.000	0.627	0.543
HSI	0.552	0.590	0.627	1.000	0.563
NIFTY	0.568	0.549	0.543	0.563	1.000

3.5 Diebold–Yilmaz results

In this step, we compute volatility connectedness using the Diebold–Yilmaz framework. Starting from the estimated VAR(p), we obtain its MA representation and then compute the generalized forecast error variance decomposition (GFEVD).

Table 11 reports GFEVD for H=1 horizon (week). Diagonal elements represent own contributions (the share explained by a market's own shocks), while off-diagonal elements capture cross-market spillovers. This matrix is the input used to construct total connectedness, directional spillovers (TO and FROM) indicies .

Table 11: Generalized FEVD matrix at H=1

	SP500	ESTOXX50	NIKKEI	HSI	NIFTY
SP500	0.385	0.210	0.164	0.117	0.124
ESTOXX50	0.199	0.365	0.198	0.127	0.110
NIKKEI	0.161	0.204	0.376	0.148	0.111
HSI	0.129	0.147	0.167	0.423	0.134
NIFTY	0.144	0.135	0.132	0.142	0.447

Total connectedness (TCI) Total connectedness (TCI) is a simple measure of how "linked" the markets are in terms of volatility. TCI is based on the off-diagonal (cross-market) shares. It tells us how much volatility in one market is explained by shocks coming from the other markets. If TCI were close to 0, each market would behave alone, volatility changes would mostly come from domestic, and shocks would not spread much across countries. If TCI were close to 100, markets would behave like a single system: a shock in one market would quickly show up in the volatility of the others.

The Figure 2 plots the TCI over time. In crisi episodes the index increases this meaning markets move more together and volatility spreads more easily across countries.

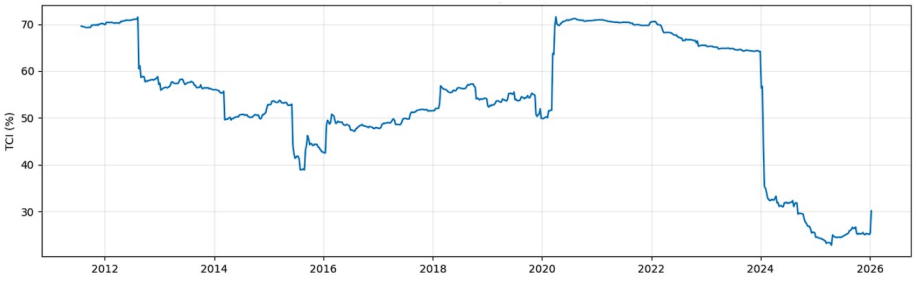


Figure 2: Total Connectedness Index (TCI) over time

Directional connectedness FROM and TO Directional measures summarize how volatility shocks propagate across markets. For each market i, $FROM_i$ measures how much of i's forecast uncertainty is driven by shocks coming from the other markets, while TO_i measures how much shocks in i contribute to the forecast uncertainty of the rest of the system. Net spillovers are defined as $NET_i = TO_i - FROM_i$: positive values indicate net transmitters, and negative values indicate net receivers.

Table 12 reports directional connectedness for three horizons ($H = 1, 4, 8$ weeks). The results can be read through a simple Western–Asian split. The Western benchmarks (S&P 500 and Euro Stoxx 50) are net transmitters on average, while the Asian benchmarks display more heterogeneous roles: Japan is mildly positive, whereas China and India are net receivers.

The horizon dimension strengthens this interpretation. As H increases, spillovers become more important (FROM rises) and the net transmitting role of the US becomes more pronounced: the S&P 500 moves from a small net transmitter at $H = 1$ (NET ≈ 1.8) to a strong transmitter at $H = 4$ and $H = 8$ (NET ≈ 18.4 and ≈ 31.0). In contrast, China and India become increasingly negative as the horizon increases, consistent with these Asian markets absorbing volatility shocks from the broader system rather than propagating them.

Table 12: Directional connectedness measures across horizons.

Market	FROM	TO	NET
	61.5429	63.3541	1.8112
SP500	60.4834	78.9032	18.4198
	59.9014	90.9381	31.0368
	63.4505	69.5579	6.1074
EUROSTOXX50	66.2893	70.0414	3.7521
	68.6023	69.3662	0.7638
	62.3552	66.0330	3.6778
JAPAN_NIKKEI	65.4393	68.0317	2.5924
	67.2218	68.1046	0.8827
	57.6984	53.4629	-4.2355
CHINA_HSI	62.5621	53.0018	-9.5603
	65.7622	52.8415	-12.9207
	55.2551	47.8941	-7.3609
INDIA_NIFTY	64.2432	49.0392	-15.2040
	69.2984	49.5358	-19.7626

*F*or each cell, the three stacked values correspond to H=1,4,8

Figure 3 shows that net connectedness is not constant over time: markets switch between being net transmitters (above zero) and net receivers (below zero). The SP 500 is positive for most of the sample and becomes especially strong around major global stress periods, suggesting that volatility shocks often originate in the US and then spread to other markets. Euro Stoxx 50 and Japan are closer to neutral on average but turn into transmitters in some subperiods. Overall, the figure supports the idea that volatility transmission between Western and Asian markets is time-varying and becomes more pronounced during crisis episodes.

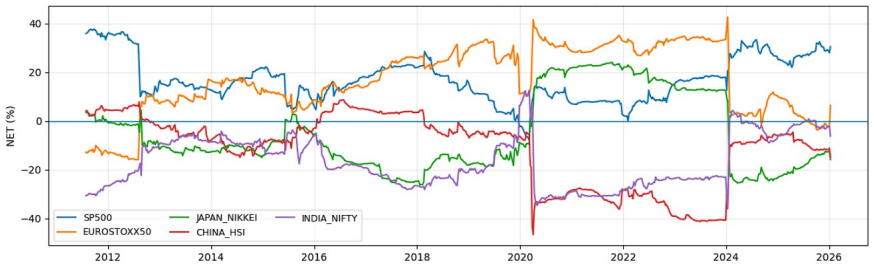


Figure 3: Net connectedness over time. Positive values indicate net transmitters.

We compute time-varying connectedness using a rolling-window VAR: at each date we re-estimate the VAR on the previous W=200 weekly observations, compute the generalized FEVD at horizon H=4

3.6 Conclusions

We started the project by hypothesizing that volatility links between Western and Asian equity markets strengthen during crisis periods, and that the SP 500 is the main transmitter of volatility.

The results largely confirm this view: During stress episodes, connectedness rises and shocks propagate more strongly across countrys. Looking at direction and net spillovers, the Western benchmarks tend to act as net transmitters on average, while the Asian markets are more mixed. Japan is close to neutral, whereas China and India are typically net receivers, this pattern becomes clearer at longer horizons, where the net transmitting role of the SP 500 increases, while China and India remain negative.

These connectedness measures can also be reused as practical indicators, for example as variables to track systemic risk, when the Total Connectedness Index (TCI) rises, volatility is increasingly driven by cross-market spillovers and diversification benefits tend to weaken, this information could be used as a signal to adjust portfolio risk exposures.

4 References

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